



00A
150.0.0.10

Ping 50.0.0.20



00B
150.0.0.20

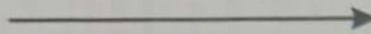
Write a packet flow of 150.0.0.10 to 150.0.0.20

ARP REQ

Sender ip	150.0.0.10
Sender mac	00A
Target ip	150.0.0.20
Target mac	

SMAC 00A

DMAC FFF

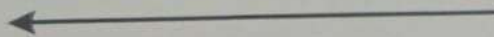


ARP REP

Sender ip	150.0.0.20
Sender mac	00B
Target ip	150.0.0.10
Target mac	00A

SMAC 00B

DMAC 00A



ECHO	SIP	150.0.0.10	SMAC	00A
	DIP	150.0.0.20		DMAC



ECHO	SIP	150.0.0.20	SMAC	00B
	DIP	150.0.0.10		DMAC



OSI LAYER (open system Interconnection)

Application

Echo

← ICMP → Internet message control protocol
32 bytes

Presentation

Echo F

→ Add format encapsulation

Session

Echo F Time

→ Add Time : encapsulation

Transport

Echo F Time X

→ packet encapsulation

~~Network~~
Network Echo F Time X Source IP: 50.0.0.10
DIP: 50.0.0.20

→ frame encapsulation

Data Link

Echo F T X
SIP: 50.0.0.10 SM: 00A
DIP: 50.0.0.20 DN: 00B

↙ Ethernet protocol

physical

(NIC - UTP)

Ethernet protocol used for add Source MAC & DMAc

ARP - Address resolution protocol

↳ It is used to find DMAc with the help of Destination IP.

1, Explain behaviour of hub device

Hub is a broadcast device, which is used to connect multiple network devices. It is a layer 1 device.

2, Explain behaviour of switch device

A switch is a layer 2 networking device, ~~that~~ which is used to connect multiple devices in a LAN. Switch is a unicast device as well as broadcast device. It uses MAC table.

3, Explain behaviour of router device

Router is a layer 3 device, which is used to connect different networks. It forwards packets based on destination IP address. It is used to break broadcast domain. It is a unicast device. It contains routing table.